Digital Twins for Real-World Applications: A Critical Analysis of Cardiovascular Healthcare

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***Abstract*—Digital Twin technology is moving out of the factory floor and marching into the hospital with a promise of a new to treat patients by creating the real virtual copy of patient's heart or body, doctors hope to predict cardiovascular events and test safe treatments before trying them out in the real world but it's proving tricky taking this industrial technology, applying it to human biology. This report reviews 15 papers in 2023 to 2025 to see whether Digital Twins can actually work for real world heart care. The analysis reveals a major three-way conflict: we need for data to be useful, we need the system to be fast, and we need patients privacy but for right now it is a hard to get all three at once. While technologies such as BlockChain and Federated Learning offer good security, they are slow at present for emergency situations where every second counts. Also, the AI that has been used in these systems often works like a black box in that it provides an answer without explaining why. Until doctors get a clear understanding of how the AI makes the decision, they cannot ethically trust it with the lives of their patients. The analysis claims that future research shouldn't be just about making the models better. Instead, concentrate on making safe and simple data connections. This includes using faster and easier encryption methods that keep the system safe without making it less efficient.**

***Index Terms*—Digital Twin, Cardiovascular Healthcare, Blockchain, Federated Learning, Cybersecurity, Critical Analysis.**

I. Introduction

The idea of Digital Twin (DT) was born in the aerospace industry during the late 1900s. Engineers at organizations like the NASA needed a way to monitor assets that were physically unreachable e.g. spacecraft in orbit. By creating a high fidelity virtual replica of it on Earth and feeding as real-time sensor data, they would be able to simulate wear and tear, predict failures, and test solutions without the risk of the actual craft [1].

In the last five years, there has been an aggressive paradigm shift towards healthcare. The vision is compelling: A Personalized Digital Twin (PDT) of a patient's heart that is updated in real time using data from smartwatches, pacemakers and hospital monitors. This twin could theoretically predict a myocardial infarction time prior to the onset of symptoms, or simulate the hemodynamic effect of preoperational stent [2].

However, applying industrial logic to human biology creates unique and severe problems. Unlike a jet engine, human data is messy, private, unstructured and protected legally by frameworks such as GDPR, HIPAA. This report provides a critical analysis of the existing research landscape, specifically focusing on the disconnection between the ideal architectures proposed in academic and the realities of hospital infrastructure.

While the papers such as Roopa and Venugopal [3] and Ringeval et al. [4] list the theoretical merits of these systems, they underestimate the costs of implementation, We analyze the conflict between **Cybersecurity** and **Performance**. For example, recent literature heavily advocates for Blockchain to secure patient data. While secure, Blockchain is inherently slow, raising the question: can we afford a 0.26 second delay when monitoring a cardiac arrest? Similarly, papers suggest using deep learning AI to make decisions, but doctors are hesitant to trust AI black boxes that cannot explain their reasoning.

This report is organized as follows: Section II outlines the related work like architecture of these systems using visual evidence from recent studies. Section III provides a deep critical analysis of the security and privacy trade-offs. Section IV future research directions. Section V concludes the study.

II. Related Work

To understand the analysis, we must initially examine the architectures presently suggested in the literature. A Digital Twin in healthcare is more than one piece of software; it is a whole system of devices, pipelines for data, and processing units.

A. *The Architecture of Cyber-Physical Systems*

Roopa and Venugopal [3] define the Digital Twin as a Cyber-Physical Healthcare System (CPHS). As illustrated in Figure 1, the architecture is split into distinct layers. The bottom layer represents the Physical World. This includes the patient and the vast array of sensors attached to them. In cardiovascular applications, this could mean wearable body sensors (such as a Fitbit designed to monitor heart rate variability), medical-grade implants (such as pacemakers) and hospital machinery. The critical challenge raised as highlighted in Figure 1 is the Connectivity Sub-Layer. All these physical devices need to get the data in to the digital world using different protocols such as 5G, Wi-Fi, Bluetooth, MQTT, etc. So we get into that very complex dependency of having to

first translate data across languages DICOM for the imaging, MQTT for IoT before it even comes to the stage for analysis.

B. Operational Applications in Nursing and Radiology

Apart from the technical architecture, the applications of Digital Twins go to the day to day operation of healthcare. Alshahranil et al. [5] performed a systematic review of these applications in nursing and radiology. The team learned that Digital Twins are being used at present in the optimization of clinical workflows and real-time prediction of patient deterioration. For example, in radiology, a Twin of the imaging department enables administrators to simulate patient flow and identify bottlenecks. However, Alshahranil et al. [5] also identified that integrating such systems requires specialized training that is presently lacking among the nursing staff, thus creating a skills gap that threatens the viability of the technology.

C. The Personalized Digital Twin (PDT)

The ultimate goal of this architecture is the creation of the Personalized Digital Twin (PDT). Shankhdhar and Garg [6] propose a framework where this twin is not just a static record, but a dynamic simulation. By using Non-Negative Matrix Factorization (NMF) and neural networks, the PDT can learn the specific normal baseline of a patient.

For example, a heart rate of 100 bpm might be normal for a patient exercising, but critical for a patient sleeping. A generic model may generate a false alarm, but a PDT that knows the habits of the patient would know the context. This context awareness is the greatest clinical value of the technology. Li et al. [7] take this a step further, proposing multi-scale twins that would work on the cellular level, predicting a reaction to a drug before a pill is ever swallowed. This is a departure from population based medicine to truly individualized care.

However, in order to create these models, massive amounts of data are required. Gross et al. [2] speak of the decentralized biobanking pathway, in which patient tissue samples are traced to form digital biological profiles. This linkage between physical samples and digital profiles is essential to the Personalized aspect of the PDT, but it also creates new vulnerabilities in this respect of who owns the digital copy of a patient's DNA.

III. Critical Analysis

A review of the selected bibliography shows a basic conflict in the design of these systems: the tradeoff between data security and system performance. This is the central Trilemma of the field; you can have Speed, Security or Utility but rarely all three at once.

A. The Centralized vs. Federated Controversy

Traditionally Digital Twins are Centralized. As shown in Figure 2, patient data is obtained from local devices (Client 1, Client 2) and uploaded to a central server (Cloud). This method is efficient in terms of computation because the central server has huge processing power. It is able to execute a complex cardiovascular simulation that would be impossible on a hospital laptop.

The Critical Flaw: Allows it simple for hackers to get in. If the central server is hacked, millions of patient records are broken into. Furthermore, transferring raw data from the hospital is often against privacy laws such as GDPR. Ewoh et al. [8] suggests that this centralized model is the major reason for the growing number of ransomware attacks on hospitals.

To solve this, Hemdan and Sayed [9] and Yang et al. [10] propose Federated Learning, illustrated in Figure 3. In this model, the patient data *never leaves the hospital*. Instead, the AI model is sent to the hospital, trains on the local data, and only sends back the mathematical updates (the Global Model updates).

Critique of Federated Learning: While Federated Learning solves the privacy problem, it introduces a **Data Quality** crisis. In a centralized system, data scientists can clean and standardize the data. They never see the raw data in a federated system. If one hospital has different calibration for its heart monitors compared to another, the AI model is learning off bad data, and the developers will not know. This lack of quality control is a dangerous gamble when precision is paramount in cardiovascular care. As shown in the cardiovascular study by Yurto glu et al. [11] large discrepancies in local data had a significant impact on the performance of the global model, unless very strict preprocessing protocols were imposed.

B. The Blockchain Bottleneck

To further secure data papers of Amofa et al. [12] and Shankhdhar & Garg [6] recommend to integrate Blockchain.They suggest that every transaction every access of a medical record every update from a pacemaker should be recorded on an immutable ledger using Smart Contracts.

The Critique: While Blockchain is a trust, it is a trust that destroys speed. Shankhdhar has obtained its own result showing a delay of 0.260 seconds per sensing instance [6].In a vacuum, this sounds fast. However, a typical ICU patient may produce thousands of data points per minute. If a system is monitoring thousands of patients, these delays are compounding, with the potential of having the Digital Twin becoming out of sync with the real-time physiology of the patient.

If a patient goes into cardiac arrest, the Digital Twin must alert the staff within the milliseconds. If the system is occupied in verifying blockchain hashes then that alert is delayed. This latency is the single biggest barrier to a real world adoption in acute care settings. The existing literature does not sufficiently discuss how blockchain can scale to process the millions of data points produced by continuous cardiac monitoring without causing fatal lag.

C. Vulnerability and Security Risks

The growth of the Digital Twin ecosystem is an exponential growth in the attack surface for bad actors. Wang et al. [13] have thoroughly described these threats as a taxonomy illustrated in Figure 4. As an example, as shown in Figure 4, the threats are not only digital, but they are physical. 4.

- **Data Poisoning:**An attacker could input false data into the Digital Twin (e.g. spoof a normal heart rate during



Fig. 1. The Data Connectivity Architecture of a Digital Twin. This diagram highlights the Physical World layer (sensors and wearables) and the complex Connectivity Sub-Layer required to transmit data to the digital system. Note the reliance on diverse protocols like 5G, Wi-Fi, and MQTT. [Source: Roopa et al. [3]]

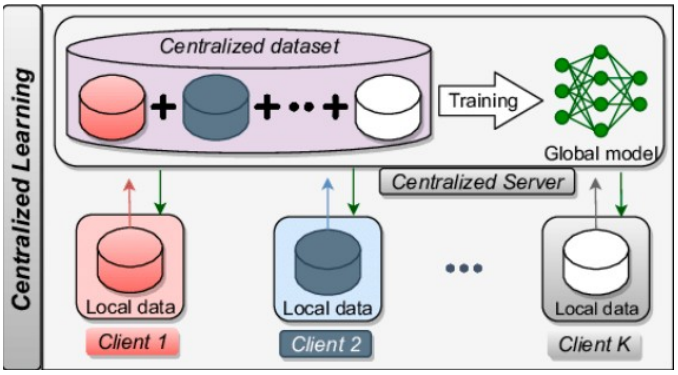


Fig. 2. Centralized Learning Architecture. Data is moved from local clients (hospitals) to a central server. This offers high computing power but creates massive privacy risks as raw data leaves the secure perimeter. [Source: Hemdan and Sayed [9]]

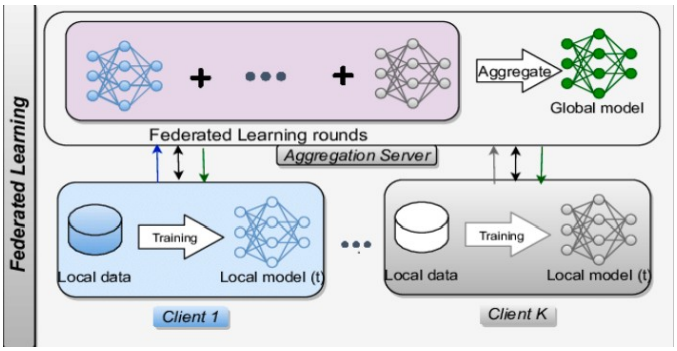


Fig. 3. Federated Learning Architecture. Instead of moving data, the AI model is distributed to the clients. Raw patient data remains local and secure, addressing privacy concerns but introducing data consistency challenges. [Source: Hemdan and Sayed [9]]

a cardiac event). This would cause the system to tell the doctor everything is fine while the patient dies. This is a specific risk in the Neural Network models proposed by Shankhdhar [6].

- **Model Inconsistency:** If the Digital Twin falls out of sync with the patient due to network failure, the doctor might make decisions based on old data. This Desynchronization is highlighted in red in Figure 4 and is a major risk identified by Wang et al. [13].
- **Ransomware:** Hospitals are prime targets for ransomware. If a Digital Twin system is encrypted by hackers, the hospital loses its ability to monitor patients effectively. Ewoh et al. [8] highlight that this is the most common sociotechnical threat facing healthcare today.

D. The Ethical Black Box and Hyper-collection

Beyond security, there is the ethical issue of Explainability. Li et al. [7] and Huang et al. [14] highlight that most AI models used in Digital Twins are Black Boxes. They give a result (e.g., Surgery Recommended) but cannot explain how they arrived at that conclusion.

In the mapping study regarding Ethical Issues by Huang et al. [14],this is what is known as Epistemic Injustice. If a Digital Twin denies a patient an insurance claim or recommends a risky procedure based upon an opaque algorithm, the patient is stripped of their agency. They cannot argue with the machine because they cannot understand it. Moreover, this lack of explanation creates a legal nightmare for doctors. If the AI is wrong, who is responsible? The doctor who followed the advice, or the Software developer who actually wrote the code? This gap is notably absent from the technical proposals of Shankhdhar [6] and Roopa [3], who focus purely on accuracy metrics.

Additionally, there is the risk of Hyper-collection of data. Papers regarding Personalized Twins (like Gross et al. [2]) often celebrate the ability to collect genetic, lifestyle, and sensor data. However, Huang et al. [14] warn that this leads to

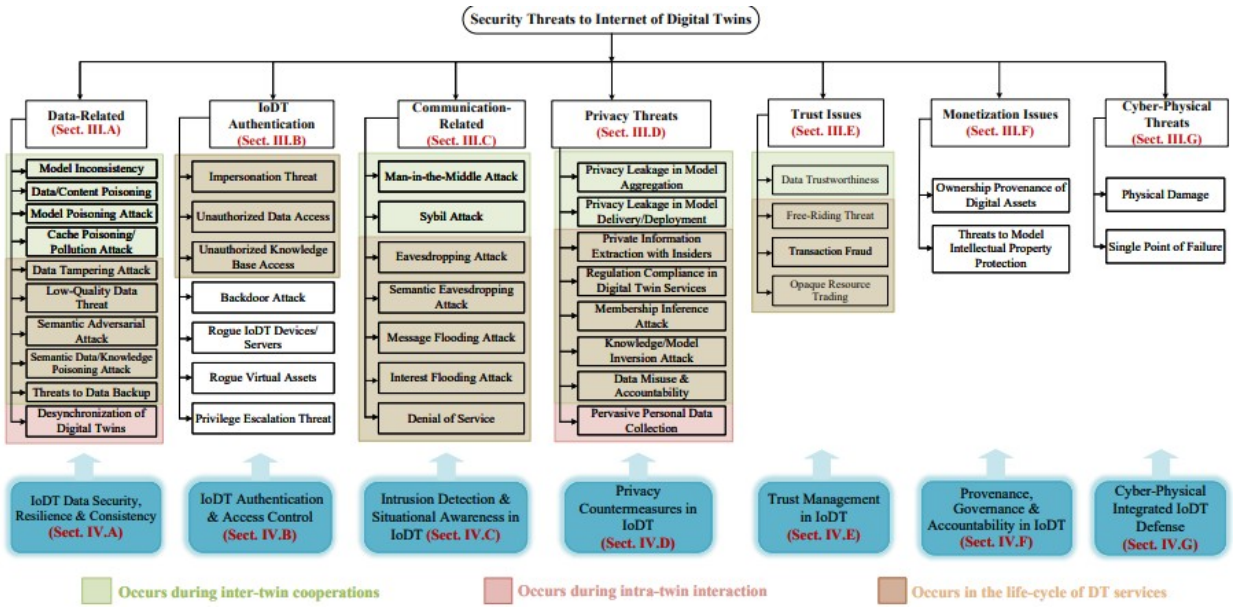


Fig. 4. Taxonomy of Security Threats to the Internet of Digital Twins (IoDT). This diagram illustrates the sheer volume of attack vectors, ranging from Data Poisoning and Model Inconsistency to physical Denial of Service (DoS) attacks. [Source: Wang et al. [13]]

Surveillance Healthcare. If a Digital Twin knows a patient ate high-sugar foods, or didn't exercise for three days, who owns that data? Can Insurance Company use the Digital Twin to deny Insurance Coverage? The technical papers tend to see more data as only beneficial, disregarding the sociological risk that patients will simply refuse to use these systems if they perceive that they are being spied upon. Iliut,a et al. [15] recommend the social factors should be included in risk assessments, in addition to the technical factors.

IV. Suggested Future Work

The current literature is rich in theory frameworks but poor in practical, scalable solutions. To bridge this gap, future research must focus on three key areas.

A. *Lightweight Security Protocols*

We must move away from standard Blockchain for IoT devices. It is too heavy. Future work must focus on Lightweight Blockchain or Directed Acyclic Graph (DAG) technologies (like the IOTA Tangle). Unlike traditional blockchain which requires mining (heavy computation), DAGs allow for faster, lighter transactions suitable for the high-speed data streams of a Digital Twin. Research is urgently needed to test if these can run on low-power devices like pacemakers without draining the battery, a concern raised by Ewoh et al. [8].

B. *Explainable AI (XAI) Integration*

Trust is the currency of health care. For doctors to adopt explanations by the AI is Digital Twins. Future work should focus on the integration of Explainable AI interfaces. Instead of simply showing a High Risk alert, the Digital Twin interface should highlight the specific variables that contribute to that risk. e.g. (Alert: Potassium levels fell 15 percent in 2 hours) This enables the doctor to check the logic of the AI and solves the Ethical dilemma set by Huang et al. [14].

C. *Resilience and Offline Mode*

Current architectures assume perfect connectivity (5G/Wi-Fi). In the real world, networks fail. Future work should create resilient twins capable of working in offline mode. The local device (edge) should be able to perform basic monitoring and alerting even in case of a lost connection to the cloud, making sure that patient safety is never compromised by a weak Wi-Fi signal. This is supported by the Sociotechnical approach espoused by Ewoh [8], with recognition of the need for technology to take account of economic growth which tends to push them to more catastrophic environmental failures.

V. Conclusion

Digital Twins has the potential to change how cardiovascular healthcare is practiced from a reactive science that treats heart attacks after they occur to a proactive one that predicts and prevents heart attacks. The literature reviewed in this report, particularly the frameworks by Shankhdhar [6], Roopa [3], and Hemdan [9],shows that the sensing and modeling technologies are mature where we can build the twin.

However, the infrastructure to support the twin is fragile. The critical analysis shows the current security proposals (like Blockchain) hurt the system performance and current AI models hurt clinical trust because they are not explainable. The proposed Personalized models will also cross ethical lines in terms of surveillance or who controls data.

The way forward then is not to create more complex biological models, but to create more robust digital infrastructures. We need security that is lightweight, transparent AI, and safe-failing systems. Only by working out these boring

infrastructural problems can the exciting promise of the Digital Twin become an everyday patient care reality.

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